Mthunzelwa Makwanyane
20241063
NET 731 PRACS – Week 6

Week 6 Practical: Cloud Security Review

Module	Week	Total Marks	NQF Level
NET731	**6 (10–16 April 2025)**	**10 Marks**	**7**

Task 1 — Security Posture Assessment (3 Marks)

Due to me needing multiple azure accounts and resources being deployed on different account there can not be a overall secure score with all the resource. Regardless, the secure score on my student account was 6%. Secure scores provide risk-based metrics to assess and monitor the cloud environment to make sure it is secure. It groups security findings into a single score for users to assess the level of danger the resource has and how to prevent it or reduce the likelihood of it affecting the azure environment. I have 17 unevaluated risks recommendations.

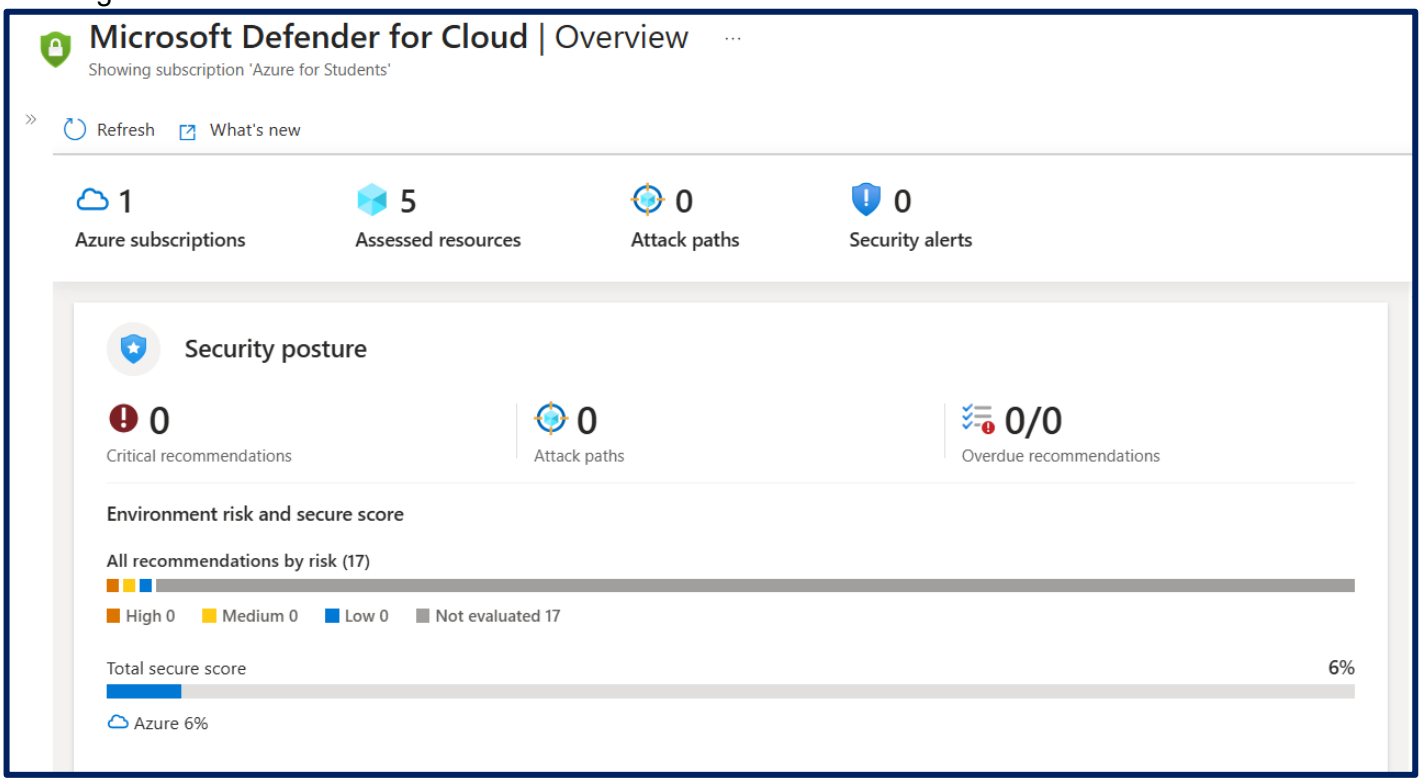

Figure 1: Display of overview and secure score

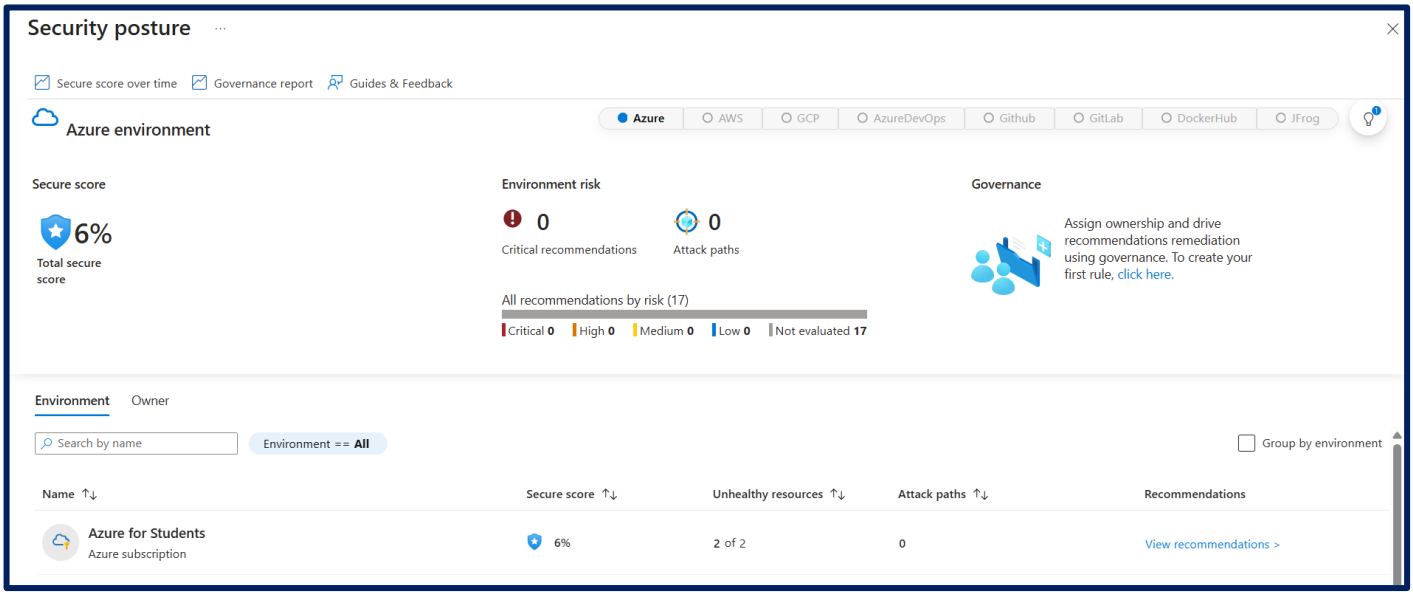

Figure 2: In-depth view of secure score

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Task 2 — Remediation (4 Marks)

2a — Close Open Management Ports (2 Marks)

In azure, to check if my environment has exposure to the public internet through port, I navigate to the Microsoft Defender for Cloud and navigate to the Attack path analysis. Attack path analysis provides a visual representation of how an attacker could reach the environment resources by highlighting the virtual machines with RDP or SSH that are exposed to the internet. For the new NSG rule, I made the source destination the IP address of the VM created in week 3 instead of setting the source to any source on the internet. Since my environment was safe a secure, the attack path analysis will stay the same

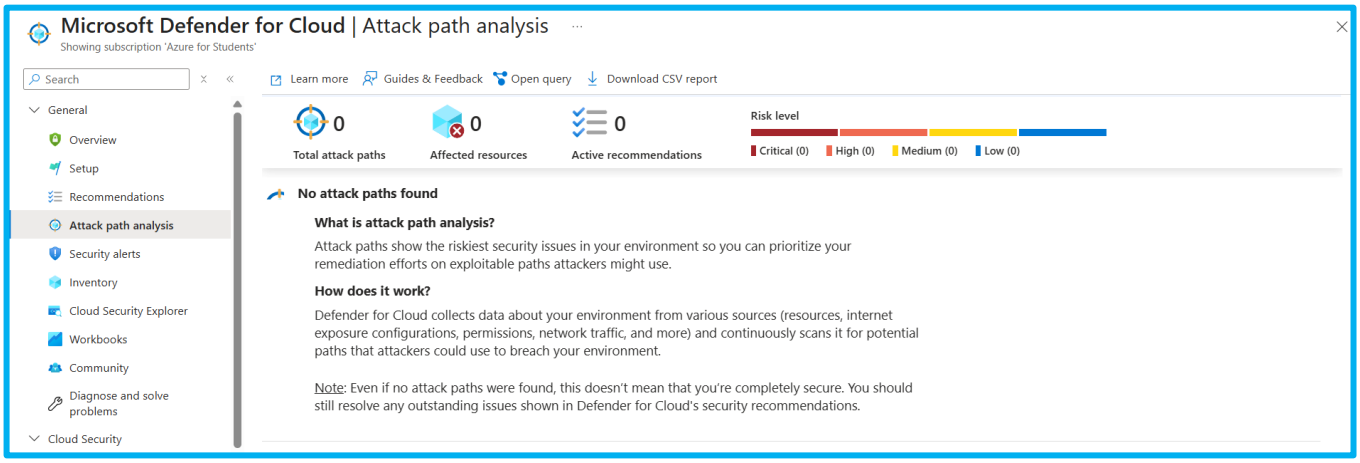

Figure 3: Attack path analysis before the NSG rule addition

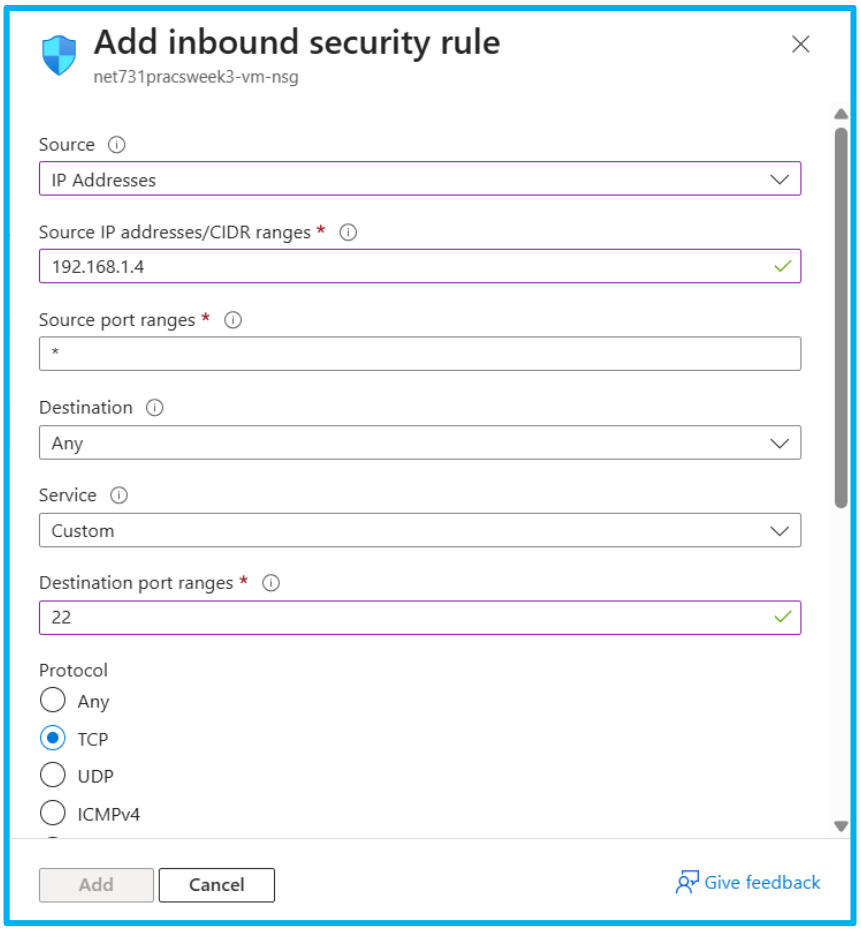

Figure 4: New NSG rule part 1

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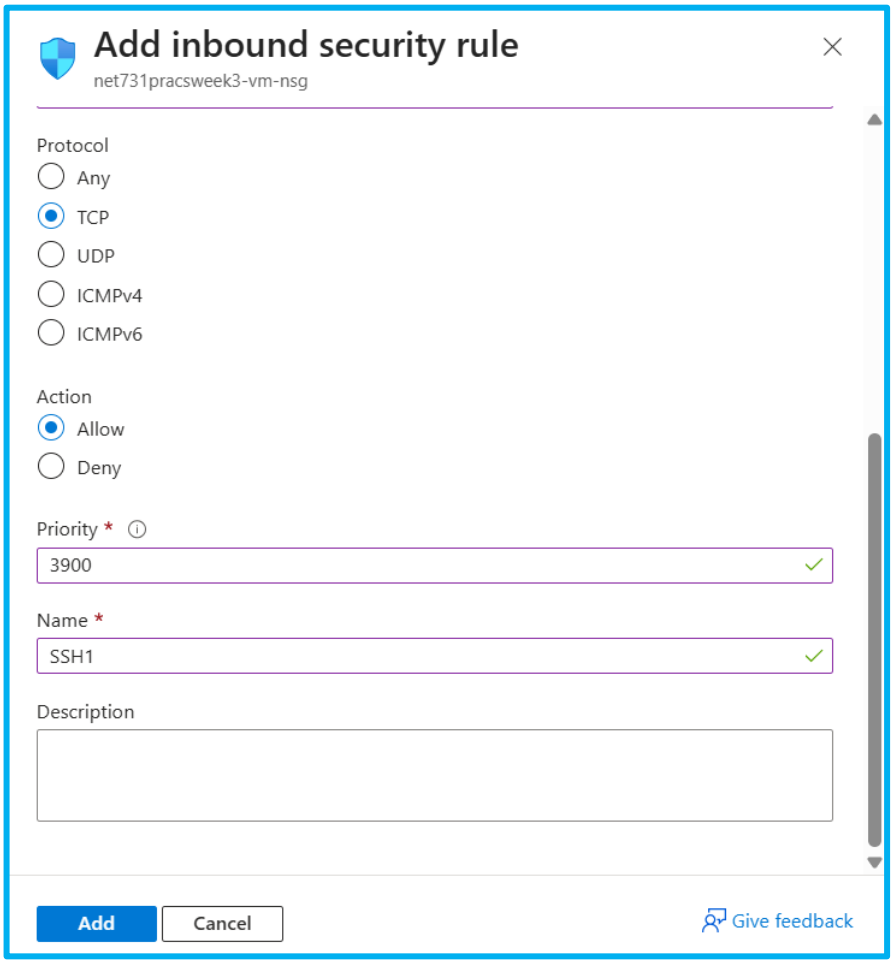

Figure 5: New NSG rule part 2

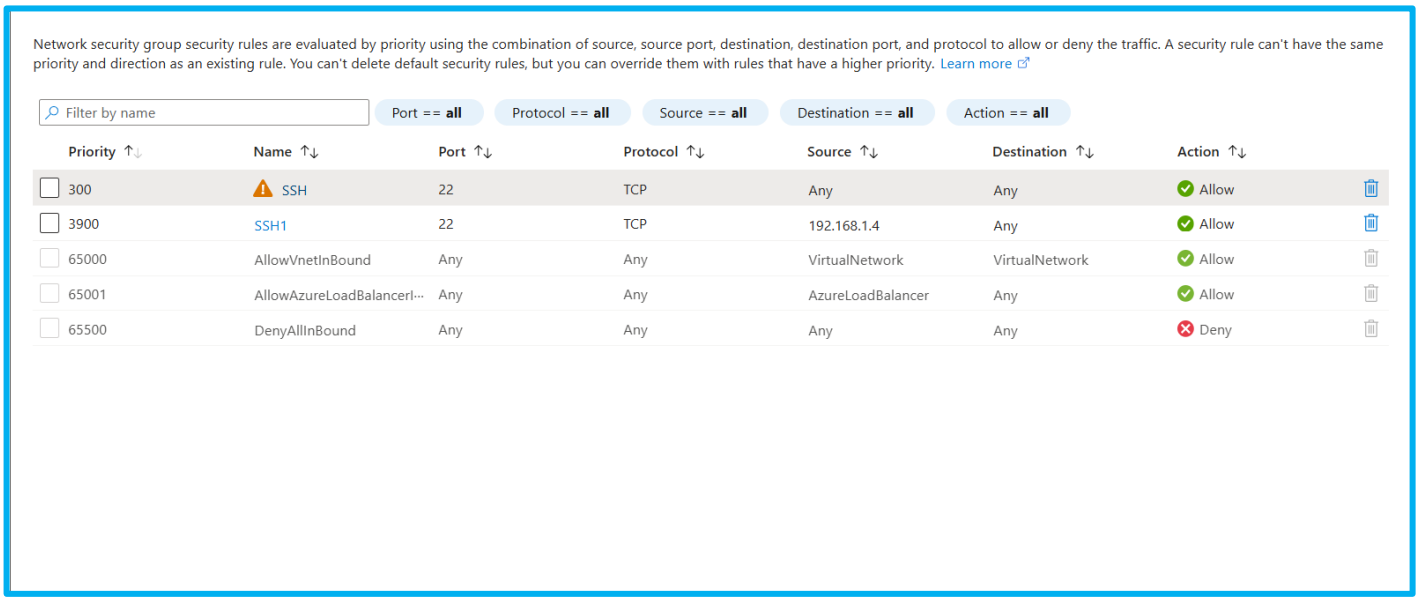

Figure 6: New SSH port NSG rule

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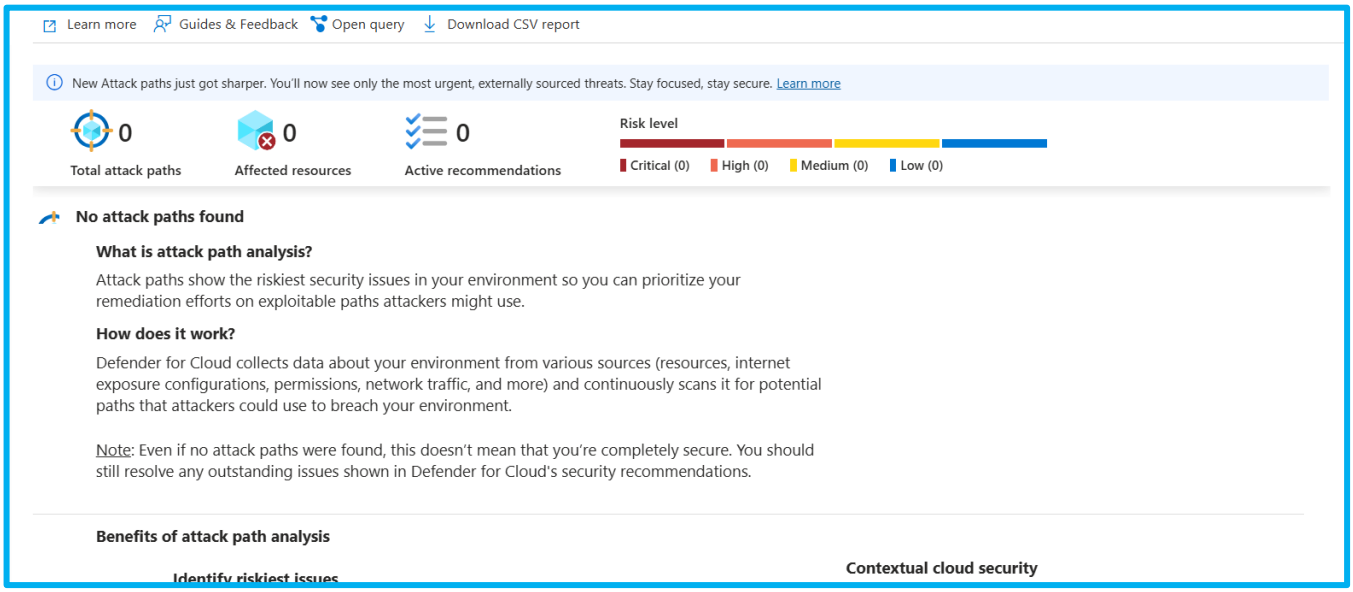

Figure 7: Attack path analysis after the new NSG rule

2b — Enable OS Updates (2 Marks)

In this section I implemented a scheduled update for my virtual machine on the 11th of May, the ending date being the 12th

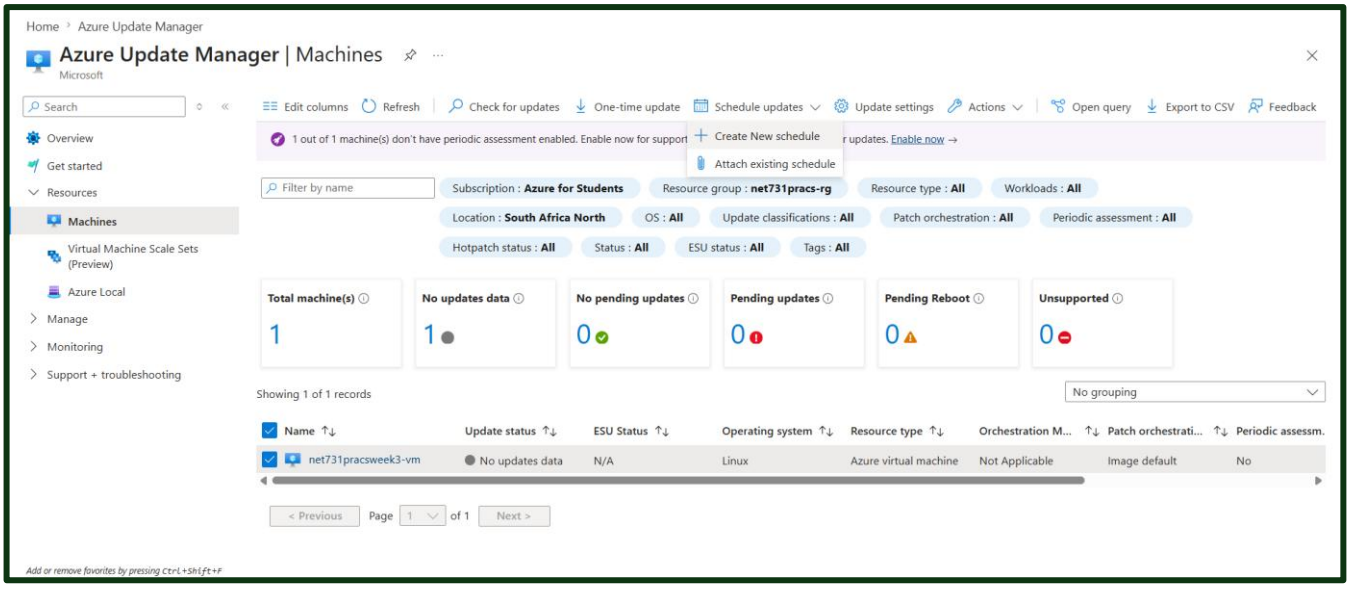

Figure 8: Creating a new schedule for my virtual machine

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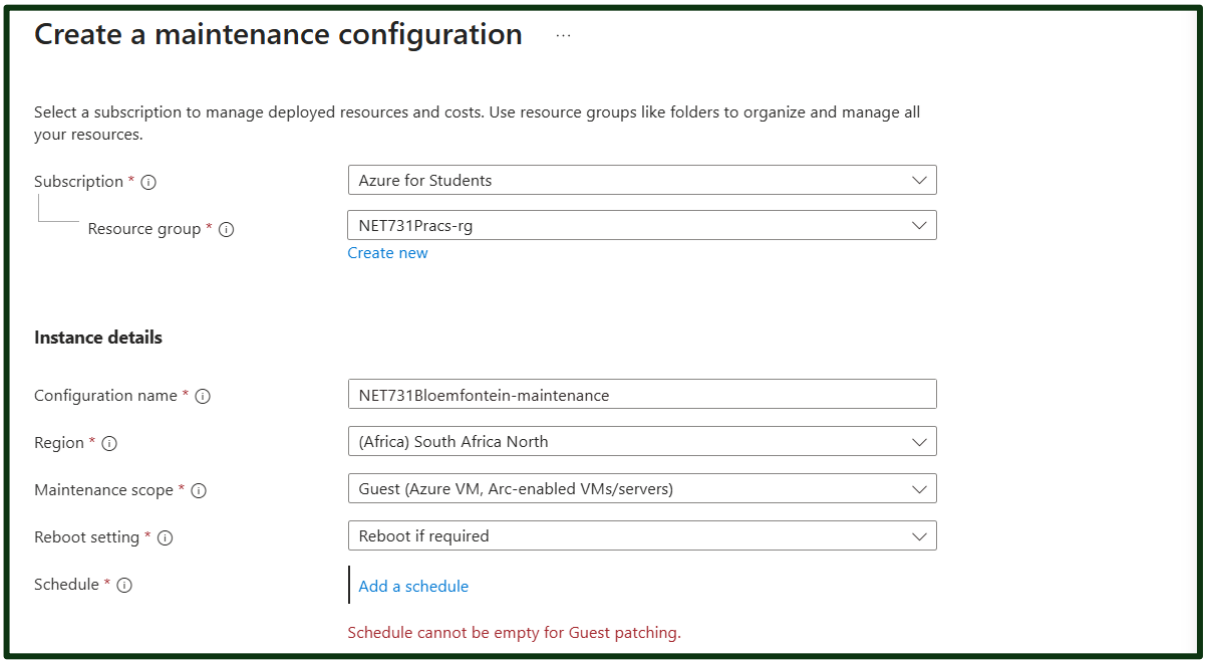

Figure 9: Entering the name, resources and region for the deployment

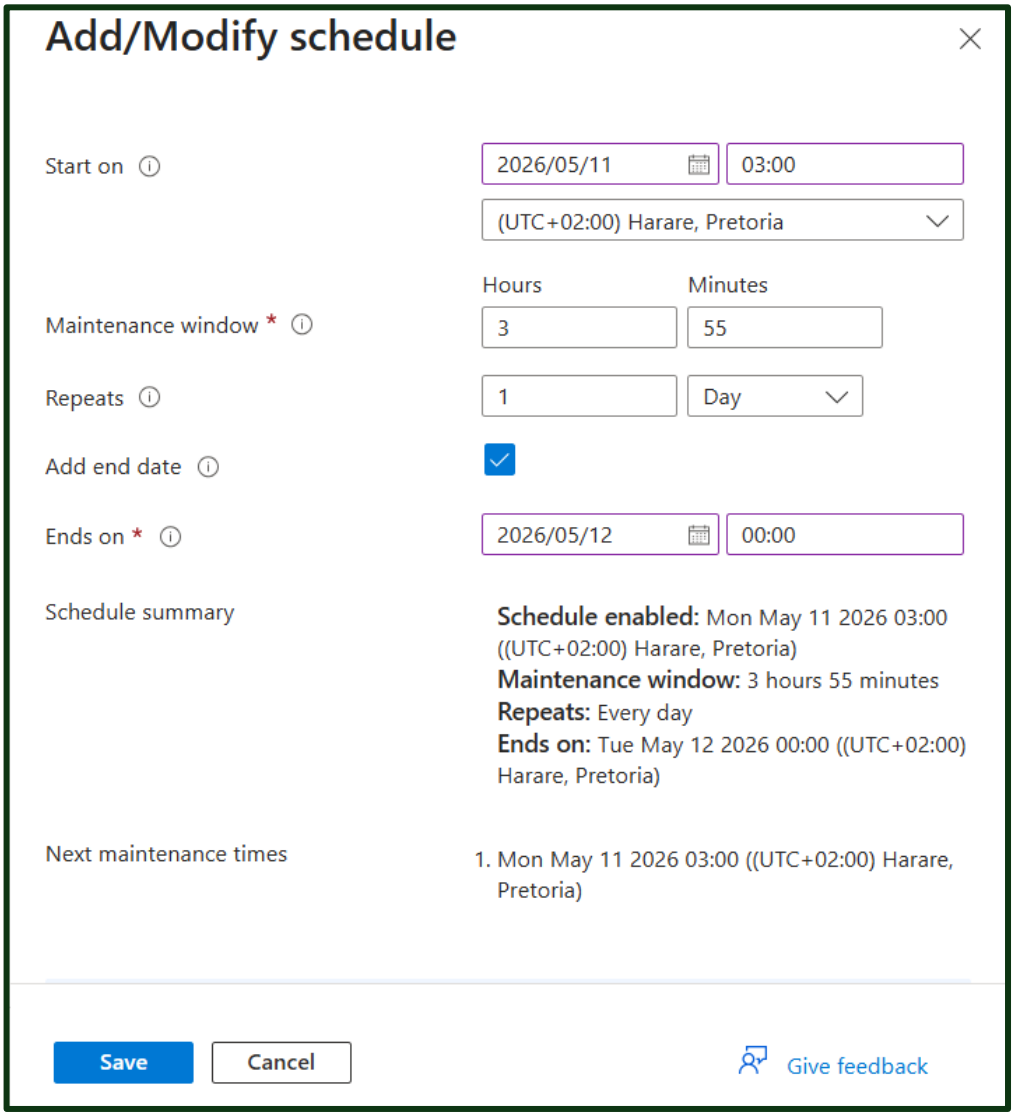

Figure 10: Adding the scheduled date for the automatic update

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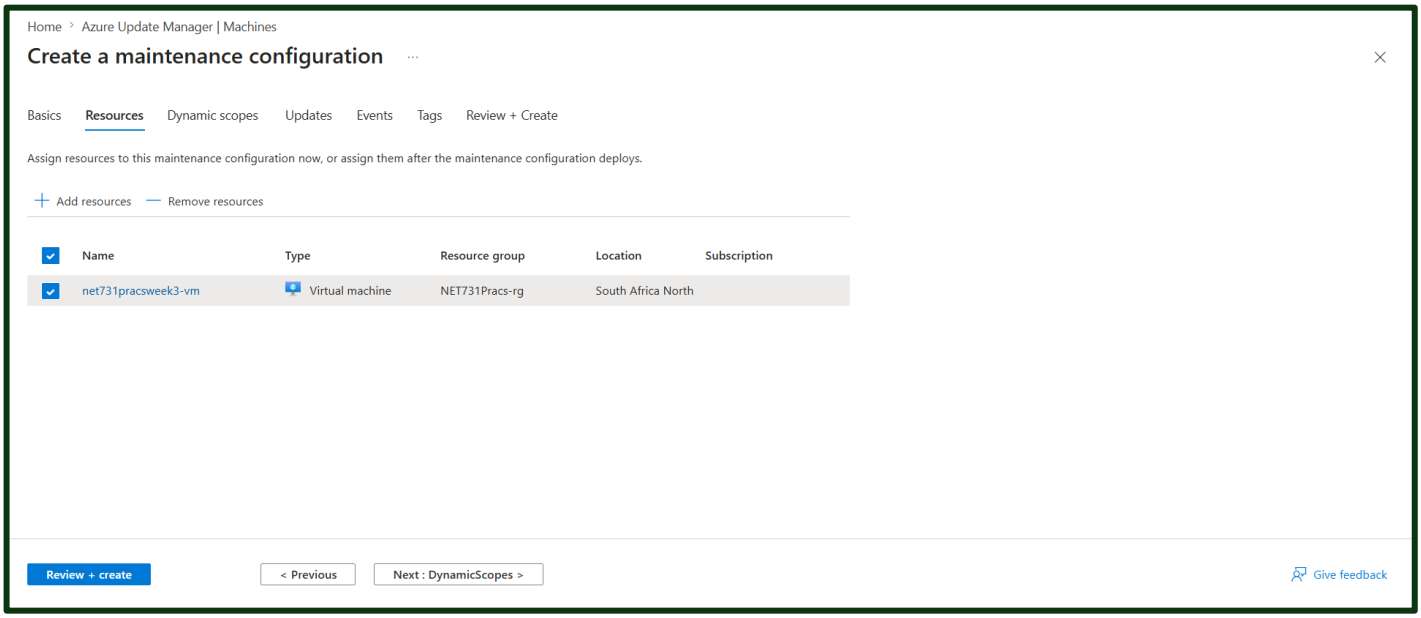

Figure 11: Selecting the desired resource for the automatic update, which is the VM

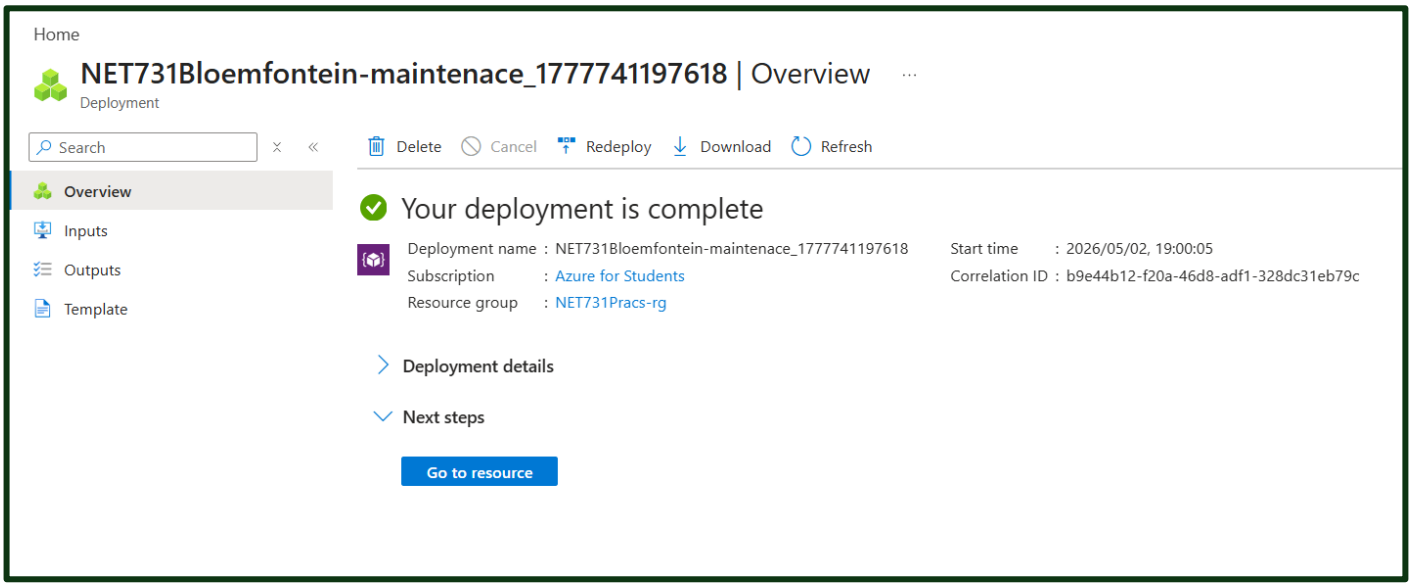

Figure 12: Proof of successful deployment

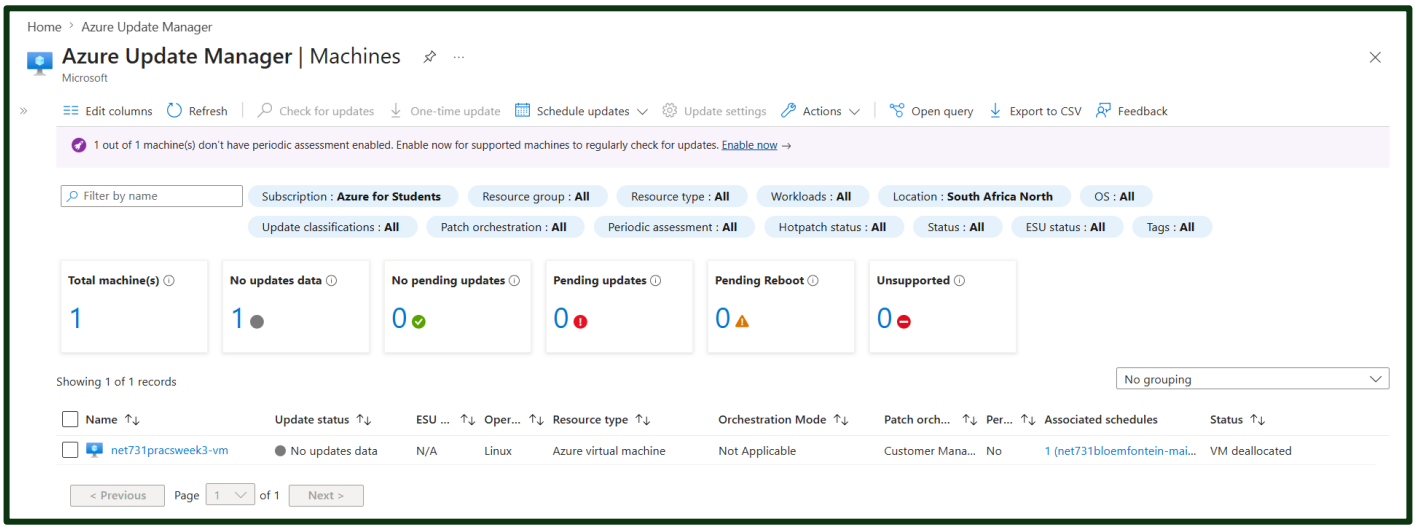

Figure 13: Proof of Scheduled update implemented on the virtual machine

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Mark Allocation

Task	Criteria	Marks
Defender for Cloud — Recommendations Review	Screenshot of Secure Score + list of top 3 active recommendations with descriptions	3
Remediation — Close open ports	Evidence of NSG rule change + before/after screenshots	2
Remediation — Enable OS updates	Screenshot of Update Manager configuration applied	2
GitHub Documentation	README with structured before/after evidence and commit history visible	3
	TOTAL	**10**